



ECE/CS/ME 539
Introduction to Artificial Neural Networks

Advanced Topics: Diffusion for Image Generation



Image Generation

Goal: Learn a model capable of generating diverse images from a base distribution of images.



Training
$$\max_{\theta} E_x [P_{\theta}(x)]$$



Generation
$$\hat{x} \sim P_{\theta}(x)$$



How to do image generation?

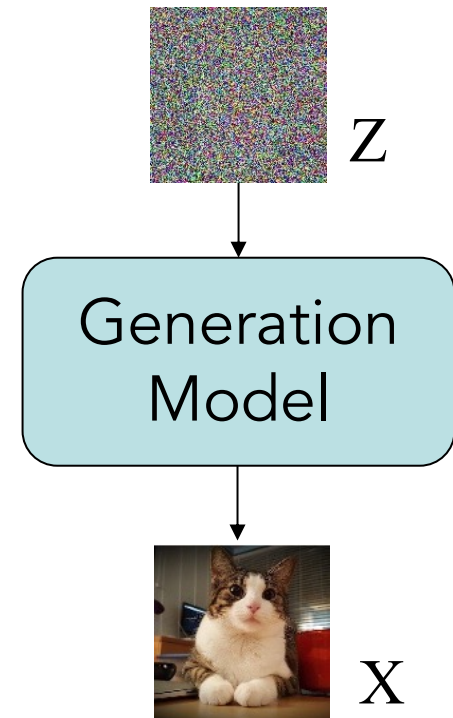
During generation, we hope to:

- Generate high quality images not present in the training set
- Generate diverse images

To accommodate the second constraint

- Randomly sample a generation seed vector:
 $z \sim N(0, 1)$
- Train a generation model f to map noise vectors $z \in Z$ (also called latents) into images $x \in X$

This idea is common over most generation models like GANs, VAEs, etc.



Problem: Transforming noise vectors into detailed images requires learning very complex functions generation models.



Diffusion Models: Progressive Denoising

Insight: Instead of seeking to transform noise into detailed images in a single step, diffusion models are trained to progressively denoise the image.

Forward diffusion process

Add noise to clean data to create increasingly noisy images

Data



Noise

Reverse denoising process

Use the noisy images to train a denoising neural network that slowly recovers the original image

[Sohl-Dickstein et al., Deep Unsupervised Learning using Nonequilibrium Thermodynamics, ICML 2015](#)

[Ho et al., Denoising Diffusion Probabilistic Models, NeurIPS 2020](#)

[Song et al., Score-Based Generative Modeling through Stochastic Differential Equations, ICLR 2021](#)

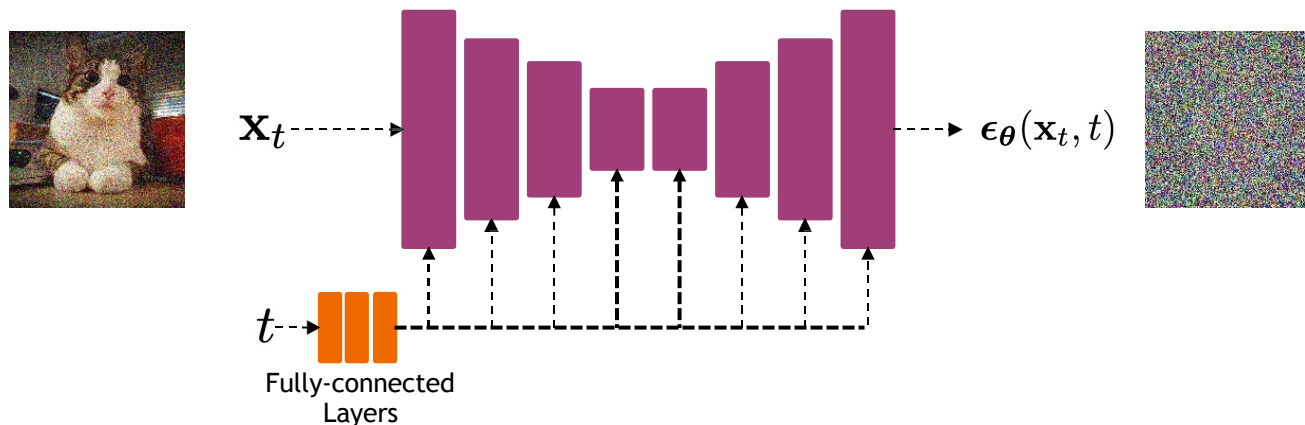


Diffusion Models: Progressive Denoising

Training strategy [Ho et al. NeurIPS 2020](#)

Let x_0 denote a training image and $\epsilon \sim \mathcal{N}(0, I)$ be pure Gaussian noise.

- Create a noisy image: $x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon$
 - α_t is the noise schedule. Higher t means more noise.
- Feed noisy image x_t and an embedding of t to a denoising network $\epsilon_\theta(\cdot)$ to predict the added random noise ϵ .
 - Loss function: $\min_{\theta} \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon - \epsilon_\theta(x_t)\|^2]$

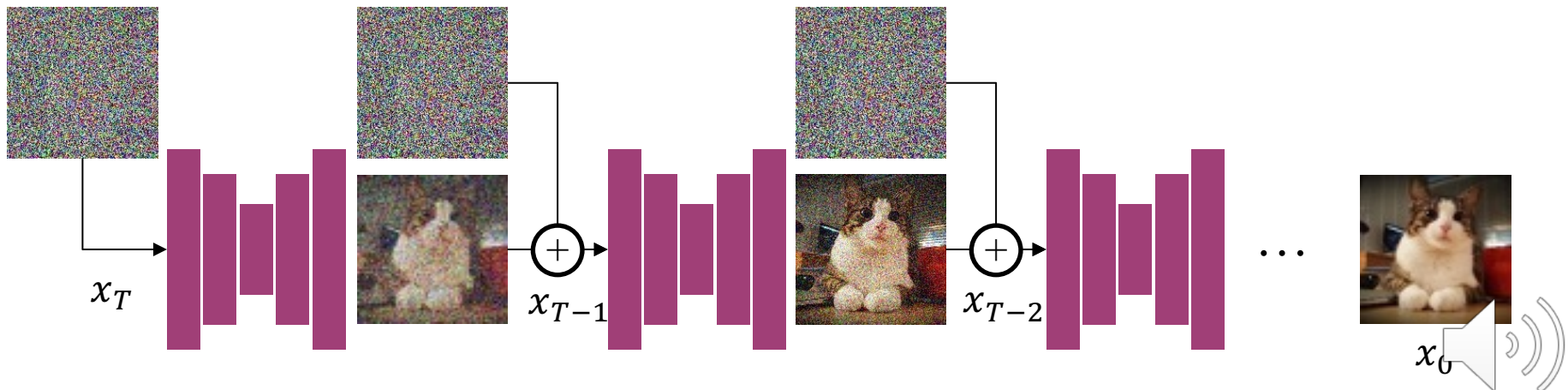


Denoising model $\epsilon_\theta(\cdot)$: U-Net with residual blocks and self-attention layers. 

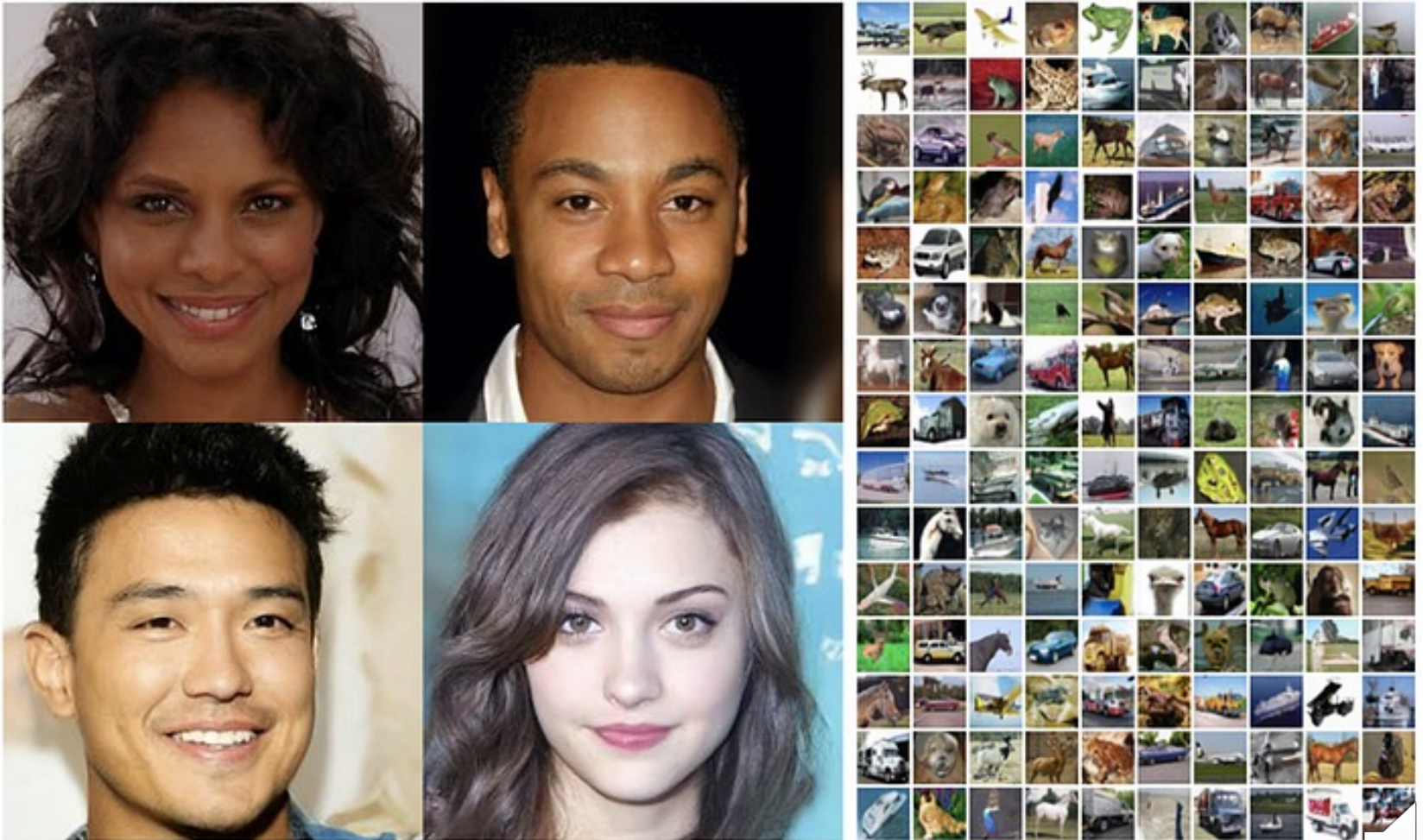
Diffusion Models: Progressive Denoising

Generation Process [Ho et al. NeurIPS 2020](#)

- Start from a pure noisy image $x_T \sim \mathcal{N}(0, I)$
- For $t = T, \dots, 1$:
 - Estimate noise in x_t using trained denoiser: $\hat{\epsilon} = \epsilon_\theta(x_t, t)$
 - Compute an imperfect estimate of image x_0 : $\hat{x}_0 = x_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \hat{\epsilon}$
 - Add noise according to noise schedule: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \hat{x}_0 + \sigma_t z$ where $z \sim \mathcal{N}(0, I)$



Diffusion works... but we have no control over what to generate



[Ho et al. NeurIPS 2020](#)

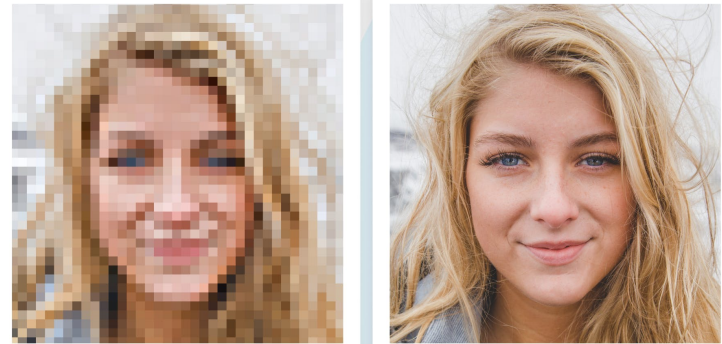
Conditional generation

Text-to-image generation

"A photo of a raccoon wearing an astronaut helmet, looking out of the window at night."



Super-Resolution



Panorama generation



← Generated

Input

Generated

Conditional diffusion models

Simply include condition as input to denoising model.

How? Depends on the type of condition.

- **Scalar conditioning** (for attributes like age, class, etc): Encode scalar into a vector embedding, which can be added to the image representations.
- **Image conditioning** (for image editing): Concatenate conditional image to generated image x_t at the input of the UNet (channel wise concatenation).
- **Text conditioning** (for text-to-image): Cross-attention layers to feed text representations to the UNet.



Imagen (Google Brain)

Text-to-image cascade diffusion

1st stage: Generates 64x64 (low res) images.

2nd and 3rd stages (super-resolution stages):

- 64x64 -> 256x256
- 256x256 -> 1024x1024

Text conditioning: Large pretrained language models as text encoders.

Training dataset: 860M image-text pairs

"A golden retriever wearing a blue barret..."

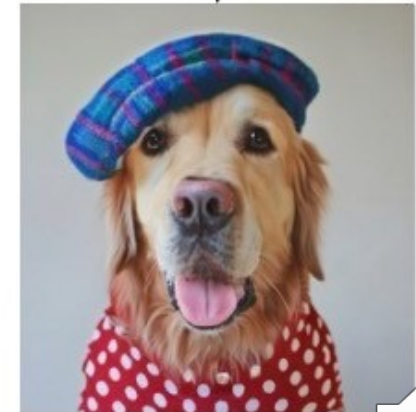


Imagen (Google Brain)



Teddy bears swimming at the Olympics 400m Butterfly event.



A cute corgi lives in a house made out of sushi.



A cute sloth holding a small treasure chest. A bright golden glow is coming from the chest.



A brain riding a rocketship heading towards the moon.



A dragon fruit wearing karate belt in the snow.



A strawberry mug filled with white sesame seeds. The mug is floating in a dark chocolate sea.



Applications

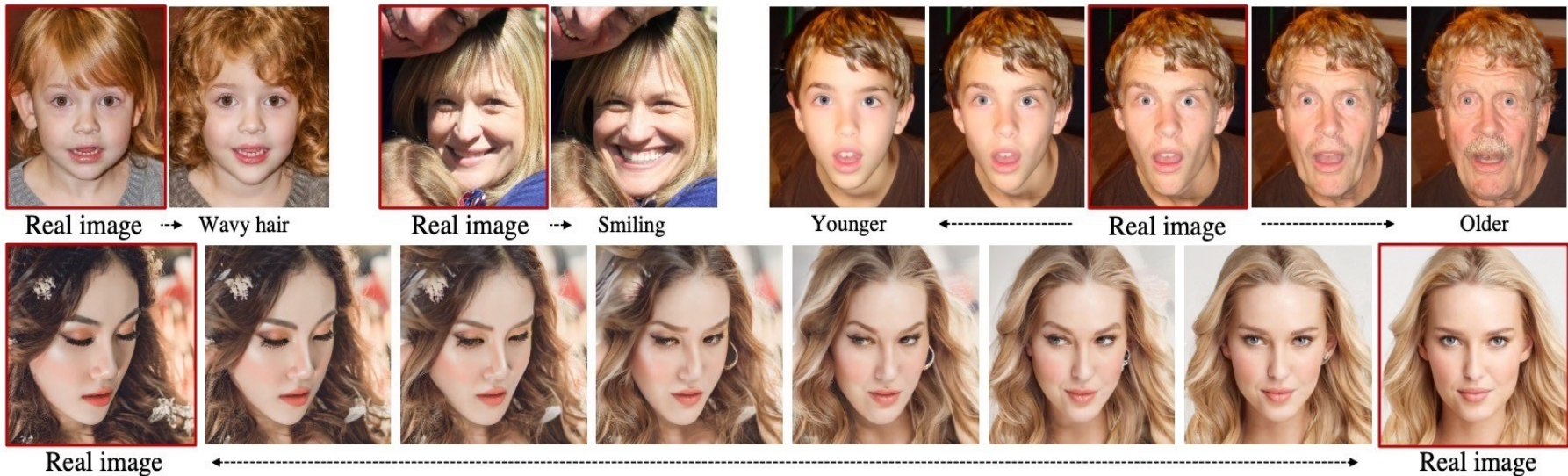
Image Editing, Image-to-Image,
Super-resolution, Segmentation



Diffusion Autoencoders

Learning semantic meaningful latent representations in diffusion models

[Preechakul et al., "Diffusion Autoencoders: Toward a Meaningful and Decodable Representation", CVPR 2022.](#)



Super-Resolution

Super-Resolution via Repeated Refinement (SR3)

[Saharia et al., Image Super-Resolution via Iterative Refinement, 2021](#)

Natural Image Super-Resolution $64 \times 64 \rightarrow 256 \times 256$



Image-to-Image Translation

Palette: Image-to-Image Diffusion Models

[Saharia et al., Palette: Image-to-Image Diffusion Models, 2022](#)

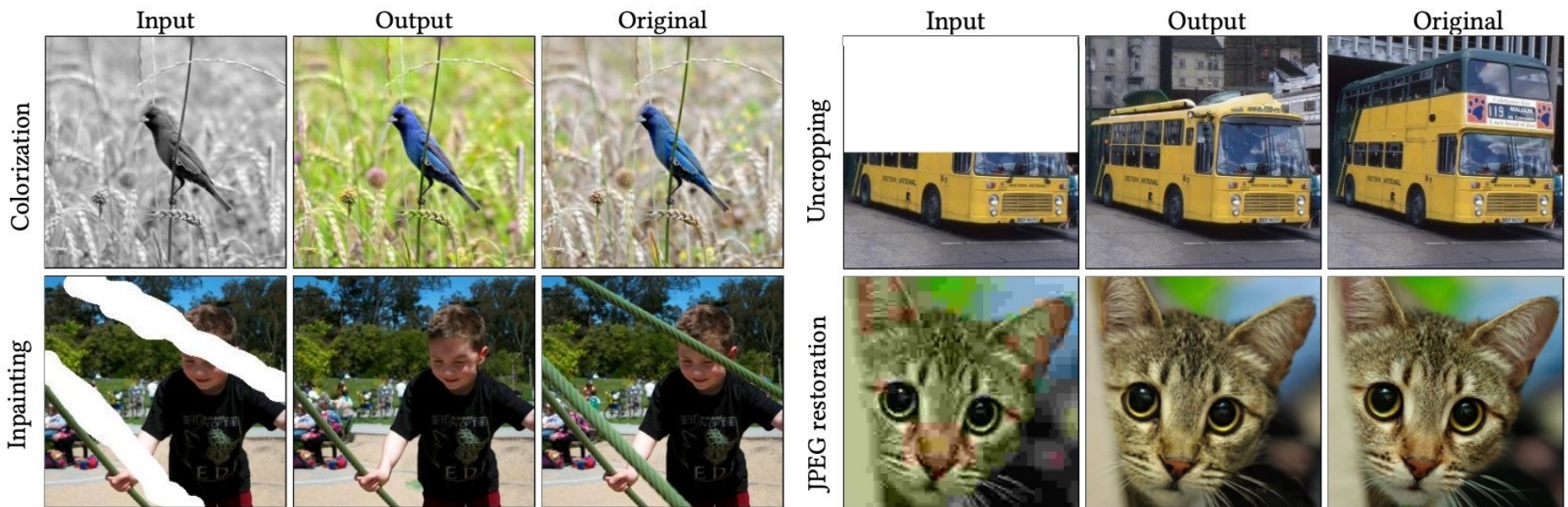
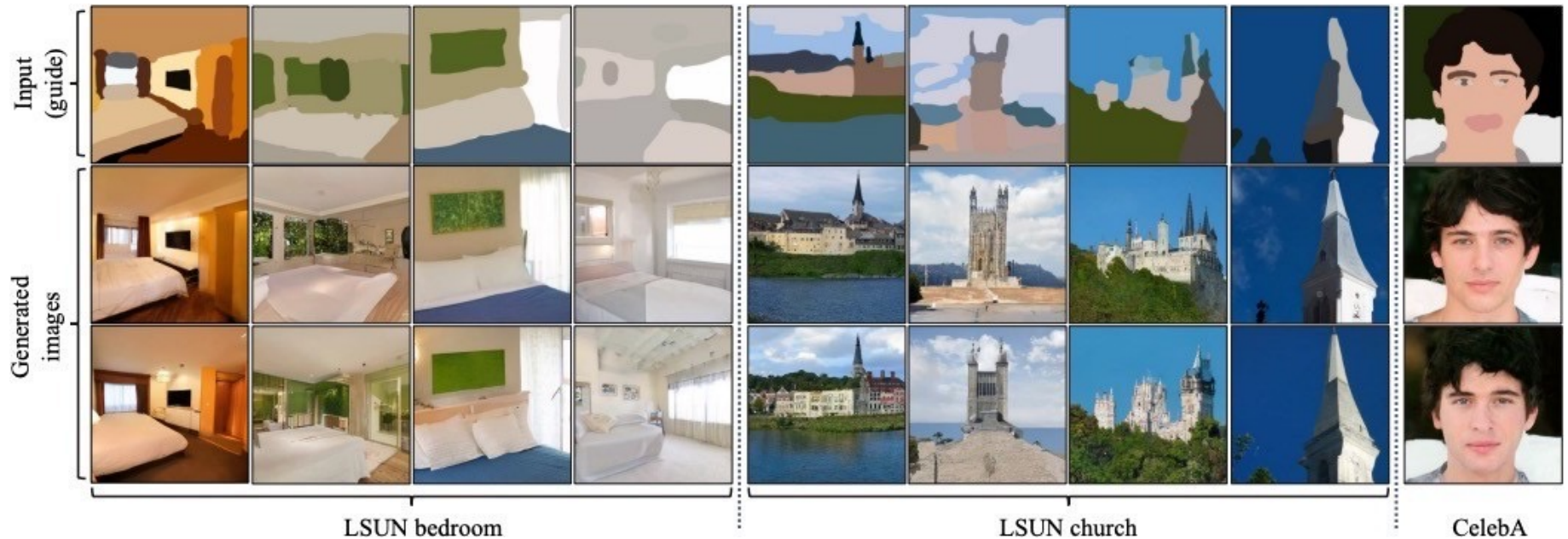


Image Editing

SDEdit

[Meng et al., SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations, ICLR 2022](#)



Summary: Diffusion Models

[Image credit: Ben Poole, Mohammad Norouzi]

